

Dietary intakes of mushrooms and green tea combine to reduce the risk of breast cancer in Chinese women.



To investigate effects of dietary mushrooms and joint effects of mushrooms and green tea on breast cancer, a case-control study was conducted in southeast China in 2004-2005. The incident cases were 1,009 female patients aged 20-87 years with histologically confirmed breast cancer. The 1,009 age-matched controls were healthy women randomly recruited from outpatient breast clinics. Information on frequency and quantity of dietary intake of mushrooms and tea consumption, usual diet, and lifestyle were collected by face-to-face interview using a validated and reliable questionnaire. Compared with nonconsumers, the Odds ratios (Ors) were 0.36 (95% CI = 0.25-0.51) and 0.53 (0.38-0.73) for daily intake of ≥ 10 g fresh mushrooms and ≥ 4 g dried mushrooms, based on multivariate logistic regression analysis adjusting for established and potential confounders. There were dose-response relationships with significant tests for trend ($p < 0.001$). The inverse association was found in both pre- and postmenopausal women. Compared with those who consumed neither mushrooms nor green tea, the ORs were 0.11 (0.06-0.20) and 0.18 (0.11-0.29) for daily high intake of fresh and dried mushrooms combined with consuming beverages made from ≥ 1.05 g dried green tea leaves per day. The corresponding linear trends were statistically significant for joint effect ($p < 0.001$). We conclude that higher dietary intake of mushrooms decreased breast cancer risk in pre- and postmenopausal Chinese women and an additional decreased risk of breast cancer from joint effect of mushrooms and green tea was observed. More research is warranted to examine the effects of dietary mushrooms and mechanism of joint effects of phytochemicals on breast cancer.